(3) The equation $\frac{dy}{dx} + y = 7x + 5$ is a differential equation.

(4) The equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = \sin x$ is a differential equation.

(5) The equation $e^{\frac{dy}{dx}} = \ln x, x > 0$ is a differential equation.

(6) The equation $\tan^{-1}\left(\frac{d^2y}{dx^2} + y^2 + 2x\right) = \frac{dy}{dx}$ is a differential equation.

**Definition 10.2 (Order of a differential equation)**

The **order** of a differential equation is the highest order derivative present in the differential equation.

Thus, if the highest order derivative of the unknown function y in the equation is $k^{th}$ derivative, then the order of the differential equation is $k$. Clearly $k$ must be a positive integer.

For example, $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} - 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4 = 0$ is a differential equation of order three.

**Definition 10.3 (Degree of a differential equation)**

If a differential equation is expressible in a polynomial form, then the integral power of the highest order derivative appears is called the **degree** of the differential equation

In other words, the **degree** of a differential equation is the power of the highest order derivative involved in the differential equation when the differential equation (after expressing in polynomial form) satisfies the following conditions :

(i) All of the derivatives in the equation are free from fractional powers, if any.

(ii) Highest order derivative should not be an argument of a transcendental function, trigonometric or exponential, etc. The coefficient of any term containing the highest order derivative should just be a function of $x$, $y$, or some lower order derivative but not as transcendental, trigonometric, exponential, logarithmic function of derivatives.

If one or more of the aforementioned conditions are not satisfied by the differential equation, it should be first reduced to the polynomial form in which it satisfies all of the above conditions.

**If a differential equation is not expressible to polynomial equation form having the highest order derivative as the leading term then that the degree of the differential equation is not defined.**

The determination of the degree of a given differential equation can be tricky if you are not well versed with the conditions under which the degree of the differential equation is defined. So go through the given solved examples carefully and master the technique of calculating the degree of the given differential equation just by sheer inspection!

**Examples for the calculation of degree:**

(1) Consider the differential equation $3y^2\left(\frac{dy}{dx}\right)^3 - \frac{d^2y}{dx^2} = \sin x^2$.



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